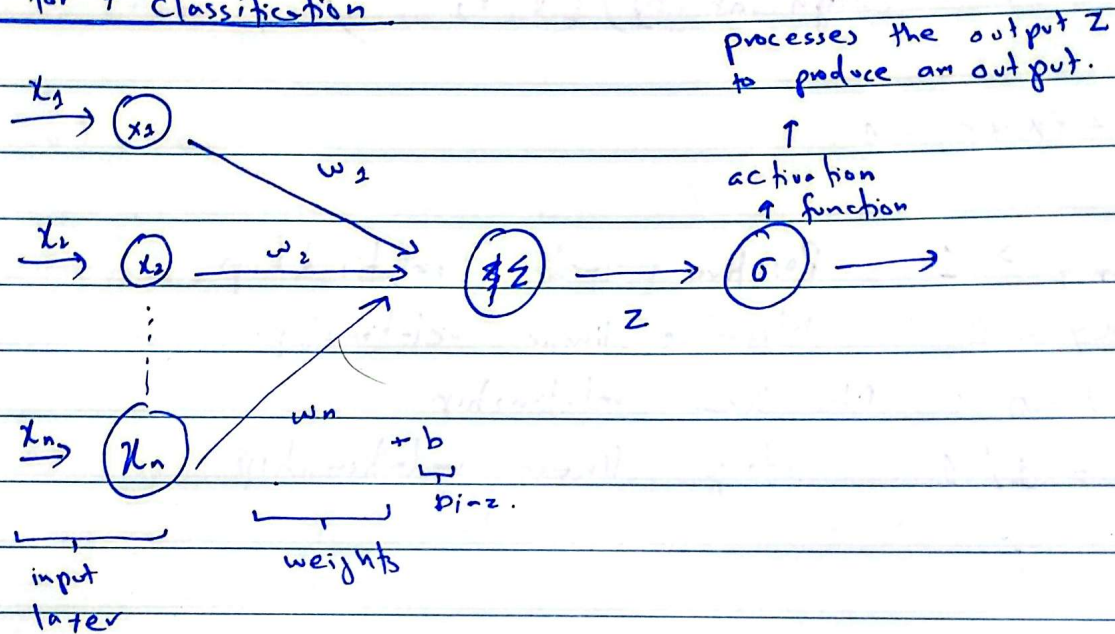


Perception Learning

Single-layer Perception

• used for ^{binary} classification



weighted sum

$$z = w_1 x_1 + w_2 x_2 + \dots + w_n x_n + b = \sum_{i=1}^n w_i \cdot x_i + b$$

$$\hat{y} = \sigma(z)$$

where σ = activation function.

Step

Step function (type of activation)

$$f(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$

where α = threshold.

α can be 0 of any value.

Perceptron learning rule

- Initialize weights and bias
- Feed the input data.
- Apply the activation function to produce an output.
- Compare the output with the actual label.
- Update the weights and bias ^{based on error} using the perceptron learning rule.

$$w_i = w_i + \Delta w_i$$

where $\Delta w_i = \alpha \cdot (y - \hat{y}) \cdot x_i$

$$b = b + \alpha \cdot (y - \hat{y})$$

where y = actual output.

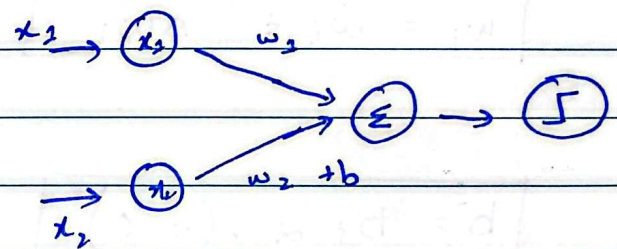
$\hat{y}$ is the predicted output.

α is the learning rate.

Example

	x_1 (Hours of study)	x_2 (Hours of sleep)	y (Pass)
1	2	5	0
2	3	6	0
3	5	8	1
4	6	7	1

initial weights : $w_1 = 0$
 $w_2 = 0$
 $b = 0$
 $\alpha = 0.1$



(using ^{binout} step activation)

$$z = w_1 x_1 + w_2 x_2 + b$$

for ①

$$z = 2(0) + 5(0) + 0 = 0 \quad y = 0$$

$$\hat{y} = 0 \quad (\text{using } \text{binout} \text{ step activation})$$

for ③

$$z = 5(0) + 8(0) + 0 = 0 \quad y = 1$$

$$\hat{y} = 0$$

$$\text{error} = \text{pred} - \text{actual} = \hat{y} - y = \boxed{-1}$$

training the perceptron (using perceptron learning rule)

$$w_i = w_i + \alpha (\hat{y} - y) \cdot x$$

$$b = b + \alpha (y - \hat{y})$$

$$w_1 = 0 + 0.1 (1 - 0) \cdot (5) = 0.5$$

$$w_2 = 0 + 0.1 (1 - 0) \cdot (8) = 0.8$$

$$b = 0 + 0.1 (1 - 0) = 0.1$$

new weights and bias:

$$w_1 = 0.5$$

$$w_2 = 0.8$$

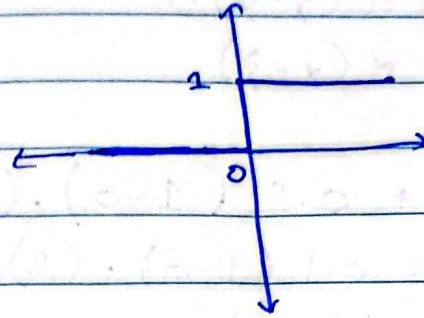
$$b = 0.1$$

~~QED~~

Common activation functions

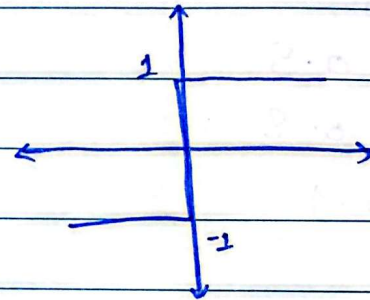
1) Binary step function

$$f(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0 \end{cases}$$



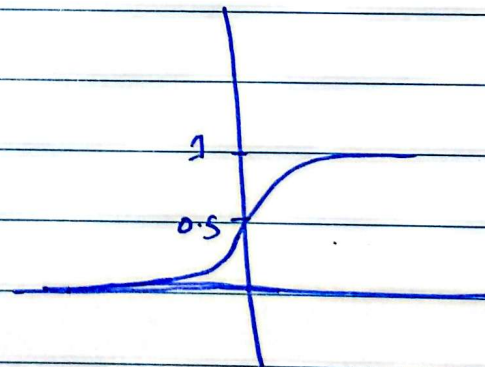
2) Bipolar step function (sign function)

$$f(x) = \begin{cases} -1 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0 \end{cases}$$



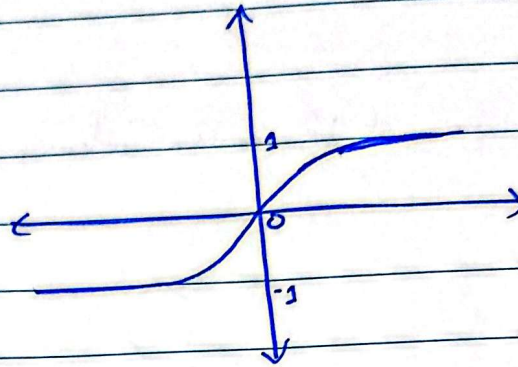
3) Sigmoid function

$$f(x) = \frac{1}{1 + e^{-x}}$$



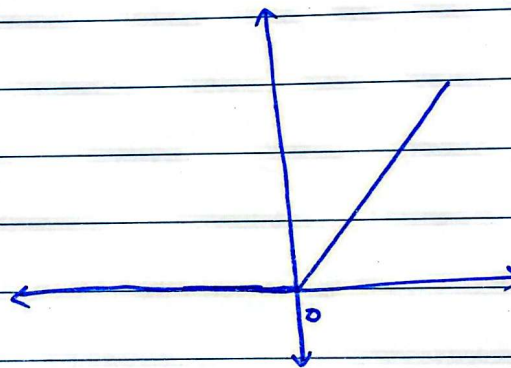
4) Tanh function

$$f(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$



5) ReLU Function.

$$f(z) = \max(0, z)$$



6) Linear activation.

$$f(z) = z$$

